Integral Trigonometri (=> kunci: Ingat segala rumus dasar Trigonometri)

1) Tipe Pangkat Genap Sin / Cos (Hal 8 no 6)

$\int \cos^4 u \, du$ => jika pangkat genap, lihat aturan halaman 5.

$\Rightarrow \int \cos^2 u \cdot \cos^{4-2} u \, du$

$\int \cos^2 u \cdot \cos^2 u \, du$

$\Downarrow$

gabung satu dulu

$\cos^n u = \cos^2 u \cos^{n-2} u$

jika n genap seperti 2, 4, 6, ...

$\cos^2 u = \dfrac{1 + \cos 2u}{2}$

$$\Rightarrow \int \cos^2 u \cdot \cos^2 u \, du = \int (\cos^2 u)^2 \, du$$

$$= \int \left(\frac{1 + \cos 2u}{2}\right)^2 du$$

$$= \int \frac{1 + 2\cos 2u + \cos^2 2u}{4} \, du$$

$$= \frac{1}{4} \int \left(1 + 2\cos 2u + \boxed{\cos^2 2u}\right) du.$$

$$= \frac{1}{4} \int \left(1 + 2\cos 2u + \left(\frac{1 + \cos 4u}{2}\right)\right) du.$$

$$= \frac{1}{4} \int \left(1 + 2\cos 2u + \frac{1}{2} + \frac{1}{2}\cos 4u\right) du.$$

$$= \frac{1}{4} \int \left(\frac{3}{2} + 2\cos 2u + \frac{1}{2}\cos 4u\right) du.$$

$$= \frac{1}{4}\left(\frac{3}{2}u + \sin 2u + \frac{1}{8}\sin 4u\right) + C.$$

$$= \frac{3}{8}u + \frac{1}{4}\sin 2u + \frac{1}{32}\sin 4u + C$$

Catatan: $\int \cos au \, du = \dfrac{1}{a}\sin au.$

sehingga $\int 2\cos 2u = 2\int \cos 2u = 2 \cdot \dfrac{1}{2}\sin 2u = \sin 2u.$

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2. Tipe Ganjil (Halaman 7 no 1) => lihat juga aturan halaman 6.

$\int \sin^m \theta \cos^n \theta \, d\theta$

$\int \sin^4 \theta \cos^5 \theta \, d\theta$ => ada pangkat genap dan ganjil ya? Pilih dulu yg ganjil.

$\int \sin^4 \theta \cdot \cos \theta \cdot \cos^{5-1} \theta \, d\theta$

$\int \sin^4 \theta \cdot \cos^4 \theta \cdot \cos \theta \, d\theta$

aturan ganjil ada juga di halaman 5

. Untuk n ganjil : $\cos^n \theta = \cos \theta \cdot \cos^{n-1} \theta$.

=> sama kek sebelumnya, namun idenya beda

=> $\int \sin^4 \theta (\cos^2 \theta)^2 \cos \theta \, d\theta$. => bentuk $\cos^2 \theta = 1 - \sin^2 \theta$.

kenapa anggap yg : $\left(\frac{1 - \cos 2\theta}{2}\right)$?

Tulis ulang lagi :

=> idenya u kan menoret ini.

$\int \sin^4 \theta \cdot \cos^4 \theta \cdot \cos \theta \, d\theta = \int \sin^4 \theta (\cos^2 \theta)^2 \cdot \cos \theta \, d\theta$

$= \int \sin^4 \theta (1 - \sin^2 \theta)^2 \cdot \cos \theta \, d\theta$.

Metode substitusi : Misalkan $u = \sin \theta$

$du = \cos \theta \, d\theta$

Sehingga $\int \sin^4 \theta (1 - \sin^2 \theta)^2 \cdot \cos \theta \, d\theta = \int u^4 (1 - u^2)^2 \cdot du$

Cepatkah dapat intuitive?? nya

$\Rightarrow \int \sin^4 \theta (1 - \sin^2 \theta)^2 \cdot \cos \theta \, d\theta = \int u^4 (1 - 2u^2 + u^4) \, du$

$= \int (u^4 - 2u^6 + u^8) \, du$

$= \frac{1}{5} u^5 - \frac{2}{7} u^7 + \frac{1}{9} u^9 + C$.

kembalikan u ke θ.

Ingat: $u = \sin \theta$.

$\frac{1}{5} \sin^5 \theta - \frac{2}{7} \sin^7 \theta + \frac{1}{9} \sin^9 \theta + C$.

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3) Tipe Menecah pangkat Csc / Sec

1) $\int \cot^2 w \csc^4 w \, dw = \int \cot^2 w \cdot \csc^2 w \cdot \csc^2 w \, dw$
$= \int \cot^2 w (\cot^2 w + 1) \csc^2 w \, dw$

Sifat : $\boxed{\csc^2 w = \cot^2 w + 1}$

Metode substitusi :

$u = \cot w$
$du = -\csc^2 w \, dw \Rightarrow -du = \csc^2 w \, dw$

$\Rightarrow \int \cot^2 w (\cot^2 w + 1) \csc^2 w \, dw = \int u^2 (u^2 + 1)(-du)$
$= -\int (u^4 + u^2) \, du$

$\Rightarrow -\left(\frac{1}{5} u^5 + \frac{1}{3} u^3\right) + c$ (Ingat : $u = \cot w$

Sehingga hasil akhir : $-\frac{1}{5} \cot^5 w - \frac{1}{3} \cot^3 w + C$.

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